

PROJECT

6.1

**VISUALIZE A
RELATIONAL DATABASE
DATABASE SERIES**

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Objective

Learn to create and manage a relational database using Amazon RDS and visualize data with Amazon QuickSight. Gain hands-on experience in connecting, securing, and analyzing cloud-based data.

This project also covers using MySQL Workbench for database tasks and building simple charts. It helps turn raw data into clear visual insights.

Services

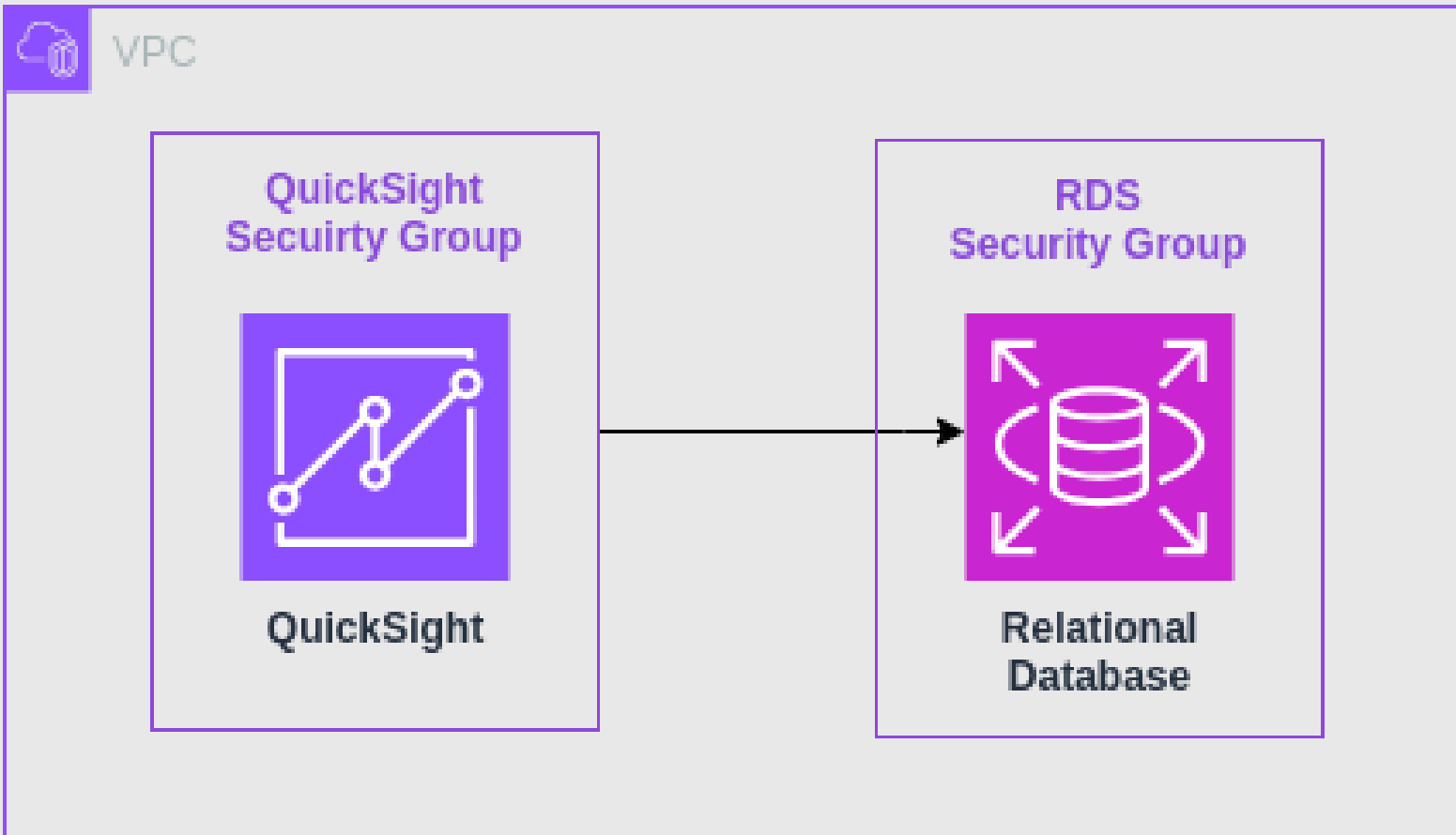
Amazon RDS - Used to create, manage, and host a scalable relational database in the cloud.

Amazon QuickSight - Enables fast, interactive dashboards and visualizations from RDS data.

IAM (Identity and Access Management) - Manages user access, roles, and ensures secure authentication.

MySQL Workbench - A GUI tool used to connect to RDS, create tables, and manage database operations.

Architecture Diagram



Amazon QuickSight and Amazon RDS are deployed within the same VPC to enable secure and private communication. They are connected using security groups that allow only authorized traffic between them, ensuring safe and efficient data visualization.

Steps to Implement 1

1. Login with IAM User:

Start by signing in to your AWS account using an IAM user with the necessary permissions. This ensures you're following best security practices. It keeps your AWS environment safe and access-controlled.

2. Create a Relational Database:

Launch an Amazon RDS instance (MySQL) inside your VPC. Choose settings like instance type, DB name, credentials, and storage. Enable public access if needed and set the right security group.

3. Connect RDS to MySQL Workbench:

Download and open MySQL Workbench on your local machine. Use the RDS endpoint, username, and password to connect. Make sure the security group allows inbound traffic from your IP.

4. Create Database Tables and Load Data:

Use MySQL Workbench to create tables using SQL commands. Insert sample records or upload CSV data into your RDS instance. This step prepares your data for visualization.

Steps to Implement 2

5. Connect RDS to QuickSight:

Go to Amazon QuickSight and choose “Manage Data” to create a new dataset. Select RDS as the source and input connection details. Choose the tables and columns you want to use.

6. Secure QuickSight Access:

Attach the necessary IAM roles to QuickSight for RDS and S3 access. Ensure QuickSight is placed in the same VPC and subnet as RDS. This enables a private and secure data flow.

7. Secure RDS and Fine-Tune Permissions:

Update the RDS security group to allow access only from QuickSight's security group or VPC. Remove any public access unless necessary. This step strengthens your database protection.

8. Reconnect and Create Visuals:

Refresh the dataset in QuickSight and reconnect if needed. Build charts like bar graphs, pie charts, or KPIs from your RDS data. Customize visuals to tell clear, meaningful stories.

Story Time

VPC (Your Cloud City):

Imagine crafting your own cloud kingdom, wrapped with secure walls that's your VPC! Inside this private city, your services like RDS and QuickSight live safely, away from the wild internet. You decide who enters or leaves through the gates called security groups.

RDS (Your Cloud Library):

Now, set up your RDS, the grand library of your cloud city, where all your data books are stored neatly on shelves. It's always ready to serve information, but only to those with the right library card (credentials and access rules). You make sure only trusted guests can walk in and read.

MySQL Workbench (Your Personal Librarian):

You bring in MySQL Workbench, your friendly librarian sitting at your desk, helping you manage and arrange all those data books. It connects securely to your cloud library and helps you organize tables and load new info all with just a few clicks and queries!

QuickSight (Your Data Artist):

Then comes QuickSight, the talented artist of your city who turns boring rows of numbers into colorful charts and stories. It visits the library, gathers the data it needs, and paints beautiful dashboards that everyone can understand. But don't worry it only enters through trusted pathways.

Security Groups (City Gates):

To keep your city safe, you place guards (security groups) at every gate. They decide who can visit the library or the artist's studio. With everyone inside the VPC and talking securely, your cloud city runs smoothly full of knowledge, beauty, and peace!

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THANK YOU!

